



RAFFLES INSTITUTION
2026 Year 6 H2 Mathematics Timed Practice Revision Paper 1
Questions and Solutions with comments

Source: 2020 Y6TP Q4		
1	Referred to the origin O, points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively such that $\mathbf{a}$ and $\mathbf{b}$ are perpendicular to each other. The point M with position vector $\mathbf{m}$ is the midpoint of A and B. It is given that $\mathbf{a} = 1$ and $\mathbf{b} = 2$.	
	(a)	Find the exact value of $ \mathbf{m} $. [3]
	(b)	Evaluate $ (2\mathbf{a} + \mathbf{b}) \times (2\mathbf{a} - \mathbf{b}) $. [3]
Solution		Comments
(a) [3]	Since M is the midpoint of A and B, $\mathbf{m} = \frac{\mathbf{a} + \mathbf{b}}{2}$. $ \mathbf{m} ^2 = \mathbf{m} \cdot \mathbf{m}$ $= \left(\frac{\mathbf{a} + \mathbf{b}}{2}\right) \cdot \left(\frac{\mathbf{a} + \mathbf{b}}{2}\right)$ $= \frac{1}{4}(\mathbf{a} \cdot \mathbf{a} + 2\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{b})$ $= \frac{1}{4}(\mathbf{a} ^2 + \mathbf{b} ^2) \quad \text{since } \mathbf{a} \cdot \mathbf{b} = 0$ $= \frac{1}{4}(1 + 4) = \frac{5}{4}$ Hence $\mathbf{m} = \frac{\sqrt{5}}{2}$.	Properties of scalar product
(b) [3]	$ (2\mathbf{a} + \mathbf{b}) \times (2\mathbf{a} - \mathbf{b}) = 4\mathbf{a} \times \mathbf{a} - 2\mathbf{a} \times \mathbf{b} + 2\mathbf{b} \times \mathbf{a} - \mathbf{b} \times \mathbf{b} $ $= 4\mathbf{b} \times \mathbf{a} \quad \text{since } \mathbf{b} \times \mathbf{a} = -\mathbf{a} \times \mathbf{b}$ $= 4 \mathbf{b} \mathbf{a} \sin 90^\circ \quad \text{or } 4 \mathbf{b} \mathbf{a} \sin 270^\circ $ $= 4 \times 1 \times 2 = 8$	Properties of vector product

Source: 2022 Y6CT Q7(a)			
2	It is given that $x \frac{dy}{dx} = y + 6xy$.		
	(i)	Using the substitution $y = v^2x$, show that the differential equation can be transformed to $\frac{dv}{dx} = f(v)$, where the function $f(v)$ is to be found. [3]	
	(ii)	Hence solve the differential equation $x \frac{dy}{dx} = y + 6xy$ to find y in terms of x . [2]	
Solution			Comments
(i) [3]	$y = v^2x$ --- (1) Differentiate w.r.t x , $\frac{dy}{dx} = 2vx \frac{dv}{dx} + v^2$ --- (2) Substitute (1) and (2) into $x \frac{dy}{dx} = y + 6xy$ $x \left(2vx \frac{dv}{dx} + v^2 \right) = v^2x + 6x(v^2x)$ $2vx^2 \frac{dv}{dx} = 6v^2x^2$ $\frac{dv}{dx} = 3v$		Some wrote $\frac{dy}{dx} = v^2$. Note that v^2 is not a constant, so there is a need to use product rule when differentiating y .
(ii) [2]	$\frac{dv}{dx} = 3v$ $\frac{1}{v} \frac{dv}{dx} = 3$ Integrate w.r.t x , $\int \frac{1}{v} dv = \int 3 dx$ $\ln v = 3x + c$ $ v = e^{3x+c}$ $v^2 = e^{6x+2c}$ $\frac{y}{x} = e^{6x+2c} \Rightarrow y = Axe^{6x}$, where $A = e^{2c}$		Most are able to write $\int \frac{1}{v} dv = \ln v + c$.

Source: 2022 Y6TP Q1		
3	(a)	Using the identity $\operatorname{cosec}^2 x = 1 + \cot^2 x$, find $\int \cot^3 x \, dx$. [3]
	(b)	Find $\int \frac{x+1}{4x^2-1} \, dx$. [3]
Solution		Comments
(a) [3]	$\begin{aligned} \int \cot^3 x \, dx &= \int \cot x \cot^2 x \, dx \\ &= \int \cot x (\operatorname{cosec}^2 x - 1) \, dx \\ &= \int \operatorname{cosec}^2 x \cot x - \cot x \, dx \\ &= -\frac{\cot^2 x}{2} - \ln \sin x + c \end{aligned}$ Alternatively, $\begin{aligned} \int \cot^3 x \, dx &= \int \cot x \cot^2 x \, dx \\ &= \int \cot x (\operatorname{cosec}^2 x - 1) \, dx \\ &= \int (\cot x \operatorname{cosec} x) \operatorname{cosec} x - \cot x \, dx \\ &= -\frac{\operatorname{cosec}^2 x}{2} - \ln \sin x + c \end{aligned}$	Recognise that we use $\int f'(x)(f(x))^n \, dx$ on , where $f(x) = \cot x$, $f'(x) = -\operatorname{cosec}^2 x$ and $n = 1$. Note that $\int \cot x \, dx = \ln \sin x + C$ and not $\ln(\sin x) + C$
(b) [3]	<u>Method 1</u> $\begin{aligned} \int \frac{x+1}{4x^2-1} \, dx &= \int \frac{1}{4x^2-1} + \frac{x}{4x^2-1} \, dx \\ &= \frac{1}{2} \int \frac{2}{(2x)^2-1} \, dx + \frac{1}{8} \int \frac{8x}{4x^2-1} \, dx \\ &= \frac{1}{2} \left(\frac{1}{2(1)} \right) \ln \left \frac{2x-1}{2x+1} \right + \frac{1}{8} \ln 4x^2-1 + c \\ &= \frac{1}{4} \ln \left \frac{2x-1}{2x+1} \right + \frac{1}{8} \ln 4x^2-1 + c \end{aligned}$ <u>Method 2</u> $\begin{aligned} \int \frac{1+x}{4x^2-1} \, dx &= \int \frac{3}{4(2x-1)} - \frac{1}{4(2x+1)} \, dx \\ &= \frac{3}{8} \int \frac{2}{(2x-1)} \, dx - \frac{1}{8} \int \frac{2}{(2x+1)} \, dx \\ &= \frac{3}{8} \ln 2x-1 - \frac{1}{8} \ln 2x+1 + c \end{aligned}$	There are quite a few who $\begin{aligned} &\int \frac{1}{4x^2-1} \, dx \\ &= \int \frac{1}{(2x)^2-1} \, dx \\ &\text{gave} \quad = \frac{1}{2(1)} \ln \left \frac{2x-1}{2x+1} \right + c \\ &= \frac{1}{2} \ln \left \frac{2x-1}{2x+1} \right + c \end{aligned}$ and forgot to divide by 2, due to the $2x$ term.

Source: 2019 Promo Q9

- 4** On 1 January 2018, Gary loans \$ A from a bank which charges compound interest at a rate of $r\%$ per month at the end of each month. Gary intends to fully repay the loan in n months with a fixed monthly instalment of \$ P_n which he pays on the first day of each subsequent month.

Show that $P_n = \frac{A(R-1)R^n}{R^n-1}$ where $R = 1 + \frac{r}{100}$. [3]

(i) Show that $\frac{P_{2n}}{P_n} = \frac{R^n}{R^n+1}$. [1]

(ii) It is given that $r = 0.55$, find the least integral number of months, n , for which the ratio $\frac{P_{2n}}{P_n}$ is greater than $\frac{3}{5}$. [2]

(iii) Show that the ratio $\frac{P_{2n}}{P_n}$ is always greater than $\frac{1}{2}$, and explain what this statement means in the context of the question. [3]

Solution

[3]	Month	Beginning of month	End of month	Given such word problems, it is best to create a table to identify the pattern.
	0 (Jan18)	A	AR	
	1 (Feb18)	$AR - P_n$	$AR^2 - P_n R$	
	2 (Mar18)	$AR^2 - P_n R - P_n$	$AR^3 - P_n R^2 - P_n R$	
	...	...	...	
	n months later	$AR^n - P_n R^{n-1} - P_n R^{n-2} - \dots - P_n$		

Since loan is fully repaid in n months,

$$AR^n - P_n R^{n-1} - P_n R^{n-2} - \dots - P_n = 0$$

$$AR^n - P_n (R^{n-1} + R^{n-2} + \dots + 1) = 0$$

$$P_n \left(\frac{1(1-R^n)}{1-R} \right) = AR^n$$

$$P_n = \frac{AR^n(1-R)}{1-R^n}$$

$$P_n = \frac{AR^n(R-1)}{R^n-1} \text{ (shown)}$$

(i) [1]	$\frac{P_{2n}}{P_n} = \frac{AR^{2n}(R-1)}{R^{2n}-1} \times \frac{R^n-1}{AR^n(R-1)}$ $= \frac{R^n(R^n-1)}{(R^{2n}-1)}$ $= \frac{R^n(R^n-1)}{(R^n-1)(R^n+1)}$ $= \frac{R^n}{R^n+1} \quad (\text{shown})$	Need to show explicitly how to factorise the denominator and cancel the terms as this is a SHOW question.
(ii) [2]	Given $r = 0.55$, to find n such that ratio is larger than $\frac{3}{5}$ $R = 1 + \frac{r}{100} = 1.0055$ $\frac{R^n}{R^n+1} > \frac{3}{5}$ $5R^n > 3R^n + 3 \quad (\text{since } R^n + 1 > 0)$ $R^n > \frac{3}{2}$ $n > \frac{\ln \frac{3}{2}}{\ln 1.0055} = 73.9$ Thus least n is 74. Alternatively, $\frac{R^n}{R^n+1} > \frac{3}{5}$ Using graphing calculator, When $n = 73$, $\frac{R^n}{R^n+1} = 0.59878 < \frac{3}{5}$ When $n = 74$, $\frac{R^n}{R^n+1} = 0.60010 > \frac{3}{5}$ Least $n = 74$	When using the GC to solve the inequality, there should be an attempt to show clearly why the least n is 74.
(iii) [3]	Since $R = 1 + \frac{r}{100} > 1$ (since $r > 0$) $R^n > 1$ $R^n + R^n > R^n + 1$ $\frac{2R^n}{R^n+1} > 1$ $\frac{R^n}{R^n+1} > \frac{1}{2}$	

	Hence $\frac{P_{2n}}{P_n}$ is always greater than $\frac{1}{2}$.	
	If Gary intends to stretch his loan term to twice as long, the monthly instalment would be more than half of what he would need to pay if he stayed on with his initial loan term.	

Source: 2022 Y6 CT Q6		
5	The molecular geometry is the three dimensional arrangement of atoms in a molecule. Fig. 1 shows the molecular geometry of a particular molecule which is of a tetrahedral form with one central atom lying inside the tetrahedron and 4 other atoms at the vertices of the tetrahedron. The points A, B, C and D represent the 4 vertices of the tetrahedron and are connected to the central atom, which is denoted by the point E, forming 4 pairs of bonding atoms AE, BE, CE and DE. Fig.1 The coordinates of A, B, C, D and E are given by $(2,3,4)$, $(0,0,3)$, $(3,0,0)$, $(0,4,0)$ and $(1,1,2)$ respectively. The plane p passes through the points A, B and C. The line l passes through the points D and E.	
	(i)	Find a cartesian equation of p . [4]
	(ii)	Find a vector equation of l and the coordinates of the point of intersection of l and p . [4]
	(iii)	Hence, find the shortest distance from point E to p . [2]
	The bond angle is the angle between any two pairs of bonding atoms, measured in degrees.	
	(iv)	Find the bond angle between bonding atoms AE and CE . [2]

Solution	Comments
(i) [4] Method 1 $\overrightarrow{AB} = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} -2 \\ -3 \\ -1 \end{pmatrix}; \quad \overrightarrow{BC} = \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ -3 \end{pmatrix}$ Since $\begin{pmatrix} -2 \\ -3 \\ -1 \end{pmatrix} \times \begin{pmatrix} 3 \\ 0 \\ -3 \end{pmatrix} = \begin{pmatrix} 9 \\ -9 \\ 9 \end{pmatrix}$, a normal vector of p is $\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$. $\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix} = 3$ $\therefore$ cartesian equation of p is $x - y + z = 3$ Method 2 Let the cartesian equation of p be $ax + by + cz + d = 0$. Since p passes through A, B and C, then $2a + 3b + 4c + d = 0$ $0a + 0b + 3c + d = 0$ $3a + 0b + 0c + d = 0$ Using GC, we get $a = -\frac{1}{3}d, b = \frac{1}{3}d, c = -\frac{1}{3}d, d \in \mathbb{R}$. Thus, $-\frac{1}{3}dx + \frac{1}{3}dy - \frac{1}{3}dz + d = 0 \Rightarrow x - y + z = 3$ (by choosing a nonzero value of d) $\therefore$ cartesian equation of p is $x - y + z = 3$	Note that the question asks for the Cartesian equation.
(ii) [4] $\overrightarrow{DE} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 0 \\ 4 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix}$. So, $l: \mathbf{r} = \begin{pmatrix} 0 \\ 4 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix}, \lambda \in \mathbb{R}$ $p: \mathbf{r} \cdot \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = 3$ When l and p intersect, $\left[\begin{pmatrix} 0 \\ 4 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} \right] \cdot \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = 3$ $-4 + 6\lambda = 3$ $\lambda = \frac{7}{6}$	There are a few who did not write the equation of the line appropriately: • Missing $\mathbf{r} =$ • Missing $\lambda \in \mathbb{R}$ Note that the question asks for the coordinates of the point of intersection.

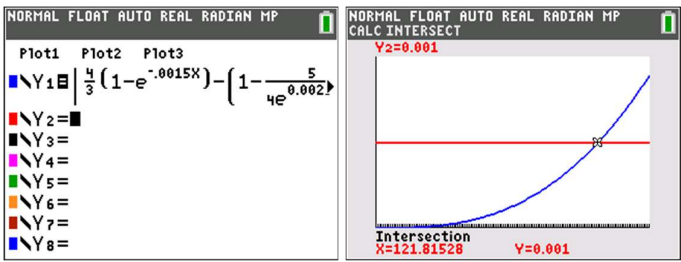
	The coordinates of the point of intersection are $\left(\frac{7}{6}, \frac{1}{2}, \frac{7}{3}\right)$	
(iii) [2]	Shortest distance $= \frac{\left \begin{bmatrix} \frac{7}{6} \\ \frac{1}{2} \\ \frac{7}{3} \end{bmatrix} - \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right }{\left \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right } = \left \begin{pmatrix} \frac{1}{6} \\ -\frac{1}{2} \\ \frac{1}{3} \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right / \sqrt{3}$ $= \left(\frac{1}{6} + \frac{1}{2} + \frac{1}{3} \right) / \sqrt{3} = \frac{1}{\sqrt{3}} \text{ units}$	Note that the question state “Hence” , so there is a need to use the previous part answer to do this part of the question.
(iv) [2]	$\overrightarrow{AE} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ -2 \end{pmatrix}$ $\overrightarrow{CE} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}$ $\cos \angle AEC = \frac{\begin{pmatrix} -1 \\ -2 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}}{\left \begin{pmatrix} -1 \\ -2 \\ -2 \end{pmatrix} \right \left \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} \right } = \frac{-4}{3(3)} = -\frac{4}{9}$ $\angle AEC = 116.4^\circ \text{ (1 d.p.)}$	It is stated in the cover page that “Give non-exact numerical answers correct to 1 decimal place in the case of angles in degrees”, some corrected the answer to 3 s.f. Quite a few did this: $\cos \angle AEC = \frac{\begin{pmatrix} -1 \\ -2 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}}{\left \begin{pmatrix} -1 \\ -2 \\ -2 \end{pmatrix} \right \left \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} \right } =$ $\angle AEC = 63.6^\circ \text{ (1 d.p.)}$ Note that the bond angle, in this question, is $\angle AEC$, which not necessarily has to be an acute angle. When finding the angle between 2 vectors, it is important to make sure that both vectors are “leaving from” or “meeting at” the same point (point E, in this question). See Chap 4B: Vectors I notes page 8.

		Thus, should use either $\overrightarrow{AE} \cdot \overrightarrow{CE}$ or $\overrightarrow{EA} \cdot \overrightarrow{EC}$, and <u>not</u> $\overrightarrow{AE} \cdot \overrightarrow{EC}$.
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Source: 2022 Y6TP Q6

6	The propagation of purchases of goods in a large population often depends on goods-related stimulus such as functionality, price and on a human tendency to imitate other people who have purchased it. The rate of increase of the proportion x, of people in a country who purchased a popular bike model named RBike, at time t weeks from the beginning of sales, can be modelled as the product of proportion of people who have not purchased it and the sum of the goods-related stimulus and imitation effects. A market analyst, who is researching on RBike sales, thinks the imitation effect per week is proportional to x with proportionality constant k known as the imitation coefficient and considers the goods-related stimulus effect as a constant $4k$ per week.
(i)	Write down a differential equation for this situation. Given that $x = 0$ when $t = 0$, solve this differential equation, giving x in terms of t and k . [5]
(ii)	Given that 10% of the population has purchased RBikes 52 weeks after sales began, show that the value of k is 0.0005 to 4 decimal places. [2]
	The market analyst's assistant wrote the differential equation incorrectly as $\frac{dx}{dt} = (1 - x)(4k) + kx$ instead, and intends to use the value of k as 0.0005 and that $x = 0$ when $t = 0$.
(iii)	Without solving this differential equation, discuss how its solution would give a different prediction of x from the solution of the differential equation formed in part (i). [1]
(iv)	Solve the differential equation written by the assistant and determine the number of weeks taken for the prediction of x to start differing by at least 0.001 between the two models. [4]

Solution	Comments
(i) [5] $\frac{dx}{dt} = (1-x)(4k+kx) = k(1-x)(4+x)$ $\frac{1}{(1-x)(4+x)} \frac{dx}{dt} = k$ Integrate w.r.t t, $\int \frac{1}{(1-x)(4+x)} dx = \int k dt$ $\frac{1}{5} \int \frac{1}{1-x} + \frac{1}{4+x} dx = \int k dt$ $-\ln 1-x + \ln 4+x = 5kt + C$ $\ln \left \frac{4+x}{1-x} \right = 5kt + C$ $\left \frac{4+x}{1-x} \right = e^{5kt+C}$ $\frac{4+x}{1-x} = \pm e^{5kt} e^C = Ae^{5kt}, \text{ where } A = \pm e^C$ When $t = 0$, $x = 0$, hence $A = 4$ We get $4+x = 4e^{5kt} - 4xe^{5kt}$ Making x the subject, we get $x = \frac{4e^{5kt} - 4}{4e^{5kt} + 1} \quad \text{OR} \quad \frac{4 - 4e^{-5kt}}{4 + e^{-5kt}} \quad \text{OR} \quad 1 - \frac{5}{4e^{5kt} + 1}$	Most are able to write down the differential equation correctly, however not many are able to solve the differential equation correctly. Note For this question, it is alright to write $\int \frac{1}{1-x} dx = -\ln(1-x) + C,$ without the modulus because $0 < x < 1$ (proportion), and so $1-x > 0$.
(ii) [2] Given that $x = 0.1$, $t = 52$, $\frac{4+0.1}{1-0.1} = 4e^{5k(52)} \Rightarrow k = \frac{1}{260} \ln \left(\frac{4.1}{4(0.9)} \right) = \frac{1}{260} \ln \left(\frac{41}{36} \right)$ $k = 0.0005002 = 0.0005 \text{ (4.dp)} \quad (\text{shown})$	As answer is given in the question, those who just wrote $\frac{4+0.1}{1-0.1} = 4e^{5k(52)}$ $\Rightarrow k = 0.0005$ will not get the mark as the marker will assume that you do not know how to solve the equation to find k, and just copy down the answer from the question.
(iii) [1] Analyst: $\frac{dx}{dt} = (1-x)(4k+kx) = (1-x)4k + kx - kx^2$ Assistant: $\frac{dx}{dt} = (1-x)(4k) + kx$ The rate of increase of x from the assistant equation is larger than the analyst equation by kx^2 since k is positive. Hence the assistant will predict a higher proportion of	

	people in the population who have purchased the RBikes as t increases, as compared to the analyst prediction. Alternative Analyst: $\frac{dx}{dt} = k(1-x)(4+x) = (1-x)(4k) + (1-x)(kx)$ Assistant: $\frac{dx}{dt} = (1-x)(4k) + kx$ Since $0 < 1-x < 1$ and $kx > 0$, the rate of increase of x given by the analyst is smaller than that given by the assistant, so the assistant model will predict a higher proportion of people in the population who have purchased the RBikes as t increases, as compared to the analyst model.	
(iv) [4]	$\frac{dx}{dt} = (1-x)(4k) + kx = 4k - 3kx$ $\frac{1}{4-3x} \frac{dx}{dt} = k$ Integrate w.r.t t, $\int \frac{1}{4-3x} dx = \int k dt$ $-\frac{1}{3} \ln 4-3x = kt + D$ $ 4-3x = e^{-3kt-3D}$ $4-3x = \pm e^{-3kt} e^{-3D} = Be^{-3kt} \text{ where } B = \pm e^{-3D}$ When $t = 0$, $x = 0$, hence $B = 4 \Rightarrow 4-3x = 4e^{-3kt}$ Since $k = 0.0005$, $x = \frac{4-4e^{-3kt}}{3} = \frac{4}{3}(1-e^{-3kt}) = \frac{4}{3}(1-e^{-0.0015t})$ Solving $\left \frac{4}{3}(1-e^{-0.0015t}) - \left(1 - \frac{5}{4e^{0.0025t} + 1}\right) \right \geq 0.001$ using GC gives $t \geq 121.815$ weeks.  Thus it takes 122 weeks for the prediction of x to start differing by at least 0.001 between the two models.	Note For this question, it is alright to write $\int \frac{1}{4-3x} dx$ $= -\frac{1}{3} \ln(4-3x) + C$ without the modulus because $0 < x < 1$ (proportion), and so $4-3x > 0$. Some wrote the inequality $\frac{4}{3}(1-e^{-0.0015t}) - \left(1 - \frac{5}{4e^{0.0025t} + 1}\right) \geq 0.001$ without the modulus. It is alright in this case because from (iii) the assistant's prediction is higher than the analyst's prediction.

Source: 2022 Y6CT Q8

7	A group of hikers is to be chosen to represent three hiking teams in an annual hiking challenge. The group is to consist of 7 hikers and is chosen at random from a set of 10 hikers consisting of 2 hikers from team A, 3 hikers from team B and 5 hikers from team C. Find the number of ways in which the group can be chosen if		
	(i)	there are no restrictions,	[1]
	(ii)	there are at least 4 hikers from team C.	[2]
	(iii)	there are at least 1 hiker from each team.	[3]

Solution			Comments																															
(i) [1]	Total selection = $^{10}C_7 = 120$																																	
(ii) [2]	Number of selections $= {}^5C_4 \times {}^5C_3 + {}^5C_5 \times {}^5C_2 = 60$ [4 hikers from Team C & 3 hikers from others + 5 hikers from Team C & 2 hikers from others]		There are a few who wrote the answer as ${}^5C_4 \times {}^6C_3 = 100$. This is not correct, as there will be many repeated counts.																															
(iii) [3]	Number of selections $= 120 - {}^8C_7 - {}^7C_7 = 111$ [No restriction – No hiker from team A – No hiker from team B] Alternatively, Number of selections $= {}^2C_1 \times {}^8C_6 + {}^8C_5 - 1 = 111$ [choose 1 out of 2 from team A & 6 hikers from others + choose both hikers from team A & 5 hikers from others – both hikers from team A are chosen & 5 hikers from team C] Alternatively, <table><tr><th colspan="3">Number of hikers from Team</th><th rowspan="2">Number of ways</th></tr><tr><th>A</th><th>B</th><th>C</th></tr><tr><td>1</td><td>1</td><td>5</td><td>${}^2C_1 \times {}^3C_1 \times {}^5C_5 = 6$</td></tr><tr><td>1</td><td>2</td><td>4</td><td>${}^2C_1 \times {}^3C_2 \times {}^5C_4 = 30$</td></tr><tr><td>1</td><td>3</td><td>3</td><td>${}^2C_1 \times {}^3C_3 \times {}^5C_3 = 20$</td></tr><tr><td>2</td><td>1</td><td>4</td><td>${}^2C_2 \times {}^3C_1 \times {}^5C_4 = 15$</td></tr><tr><td>2</td><td>2</td><td>3</td><td>${}^2C_2 \times {}^3C_2 \times {}^5C_3 = 30$</td></tr><tr><td>2</td><td>3</td><td>2</td><td>${}^2C_2 \times {}^3C_3 \times {}^5C_2 = 10$</td></tr></table> Total number of ways = $6 + 30 + 20 + 15 + 30 + 10 = 111$		Number of hikers from Team			Number of ways	A	B	C	1	1	5	${}^2C_1 \times {}^3C_1 \times {}^5C_5 = 6$	1	2	4	${}^2C_1 \times {}^3C_2 \times {}^5C_4 = 30$	1	3	3	${}^2C_1 \times {}^3C_3 \times {}^5C_3 = 20$	2	1	4	${}^2C_2 \times {}^3C_1 \times {}^5C_4 = 15$	2	2	3	${}^2C_2 \times {}^3C_2 \times {}^5C_3 = 30$	2	3	2	${}^2C_2 \times {}^3C_3 \times {}^5C_2 = 10$	
Number of hikers from Team			Number of ways																															
A	B	C																																
1	1	5	${}^2C_1 \times {}^3C_1 \times {}^5C_5 = 6$																															
1	2	4	${}^2C_1 \times {}^3C_2 \times {}^5C_4 = 30$																															
1	3	3	${}^2C_1 \times {}^3C_3 \times {}^5C_3 = 20$																															
2	1	4	${}^2C_2 \times {}^3C_1 \times {}^5C_4 = 15$																															
2	2	3	${}^2C_2 \times {}^3C_2 \times {}^5C_3 = 30$																															
2	3	2	${}^2C_2 \times {}^3C_3 \times {}^5C_2 = 10$																															

Source: 2022 Y6CT Q9

- 8** A study found that students at the National University of Singapore and the Nanyang Technological University sleep an average of 6.2 hours a day. Andy surveys 50 students from university S to estimate each person's daily sleeping hours. The sample mean is $\bar{t}$ hours and the sample standard deviation is 2.2 hours. A test is to be carried out at the 5% level of significance to draw conclusions on the average daily sleeping hours of students from university S .

Source: <https://www.straitstimes.com/singapore/lack-of-sleep-is-nothing-to-yawn-about>

- (i) Find the set of values of $\bar{t}$ for which he will conclude that the students from university S sleep more than 6.2 hours on average. State the appropriate hypotheses for the test, defining any symbols you use. [5]

- (ii) State the set of values of $\bar{t}$ for which he will conclude that the average sleeping hours of students from university S is not 6.2 hours. [1]

Solution**Comments****(i)**
[5]

An unbiased estimate for the population variance,

$$s^2 = \frac{50}{50-1}(2.2^2) = 4.93877551 \text{ hours}^2$$

Let T be the number of sleeping hours of a student from university S and let μ hours be the population mean sleeping time of the students from university S .

Null hypothesis $H_0 : \mu = 6.2$ Alternative hypothesis $H_1 : \mu > 6.2$

Under H_0 , $\bar{T} \sim N\left(6.2, \frac{4.93877551}{50}\right)$ approximately by

Central Limit Theorem, since $n = 50$ is large.

To conclude that the mean sleeping hours is more than 6.2 hours, i.e. for H_0 to be rejected,

$$p\text{-value} \leq 0.05$$

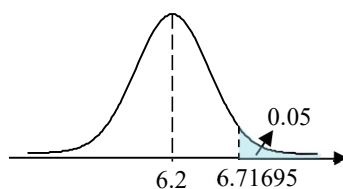
$$P(\bar{T} \geq \bar{t}) \leq 0.05$$

From GC,

$$P(\bar{T} \geq 6.71695) = 0.05$$

From the diagram, $\bar{t} \geq 6.71695$

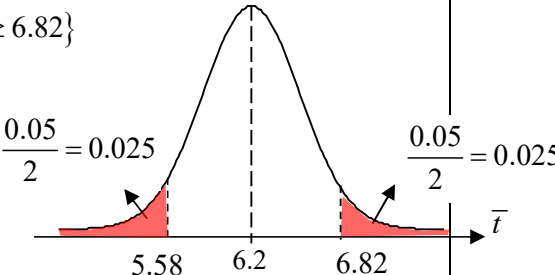
$\therefore$ the set of values of $\bar{t}$ is $\{\bar{t} \in \mathbb{R} : \bar{t} \geq 6.72\}$ (3 s.f.).



Many did not define μ properly.

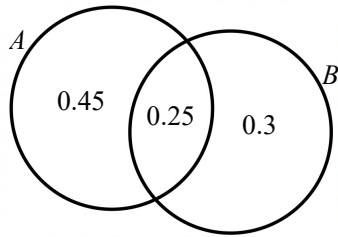
The given standard deviation in the question is the **sample** standard deviation.

Many did not find the **unbiased estimate** for the population variance.

(ii) [1]	$H_0 : \mu = 6.2$ $H_1 : \mu \neq 6.2$ To conclude that the mean sleeping hours is not 6.2 hours, i.e. for H_0 to be rejected, $\bar{t}$ must lie in the critical region. i.e. $P(\bar{T} \leq \bar{t}) \leq \frac{0.05}{2} \quad \text{or} \quad P(\bar{T} \geq \bar{t}) \leq \frac{0.05}{2}$ $\bar{t} \leq 5.5840 \quad \text{or} \quad \bar{t} \geq 6.8160 \text{ (5s.f.)}$ Therefore, the required set of values of $\bar{t}$ is $\{\bar{t} \in \mathbb{R} : \bar{t} \leq 5.58 \text{ or } \bar{t} \geq 6.82\}$ 	
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Source: 2022 Y6CT Q10

9	For events A and B , it is given that $P(A') = 0.3$ and $P(B) = 0.55$.		
	(i)	Given that the events A and B are independent, find $P(A \cap B)$. [1]	
	(ii)	Given that events A and B are not independent, find the range of $P(A \cup B)$. [2]	
	(iii)	Given that $P(A' B) = 0.4$, find	
	(a)	$P(A \cup B)$, [3]	
	(b)	$P(A' \cup B')$. [2]	
Solution			Comments
(i) [1]	$P(A \cap B) = 0.7(0.55) = 0.385$, since A and B are independent		
(ii) [2]	$P(A) = 0.7$, $P(B) = 0.55$. The least value of $P(A \cup B)$ occurs when $B \subseteq A$. Then the least value of $P(A \cup B)$ is 0.7. The largest value of $P(A \cup B)$ occurs when $P(A \cup B) = 1$. This is possible by considering $P(A \cap B) = 0.25$. (see Venn diagram below)		

	 From (i), when A and B are independent, $P(A \cap B) = 0.385$. Then, the corresponding $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $= 0.7 + 0.55 - 0.385$ $= 0.865$ Since A and B are not independent, then we need to exclude $P(A \cup B) = 0.865$. Thus, range of $P(A \cup B)$ is $[0.7, 1] \setminus \{0.865\}$.	
(iii)	Method 1 (a) $P(A' B) = 0.4 \Rightarrow P(A' \cap B) = 0.4 \times 0.55 = 0.22$ [3] Thus, $P(A \cup B) = P(A' \cap B) + P(A) = 0.22 + 0.7 = 0.92$ Method 2 $P(A B) = 1 - P(A' B) = 1 - 0.4 = 0.6$ $P(A \cap B) = P(B) P(A B) = (0.55)(0.6) = 0.33$ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $= 0.7 + 0.55 - 0.33$ $= 0.92$	
(iii) (b) [2]	$P(A' \cup B') = 1 - P(A \cap B)$ $= 1 - 0.33$ $= 0.67$	Quite a significant number of students wrote: $P(A' \cup B') = 1 - P(A \cup B)$ This is not correct, although $P(A') = 1 - P(A)$. See from the Venn diagram $P(A' \cup B') = 1 - P(A' \cap B')$

Source: 2022 Y6CT Q11

- 10** A policeman inspects each passenger in the car that he stops during a roadblock. The number of passengers in a randomly stopped car is the random variable X . The probability distribution of X is shown in the table below.

x	1	2	3	4	5
$P(X = x)$	0.32	0.43	0.16	0.07	k

- (i)** Find the value of k and determine $E(X)$. [2]

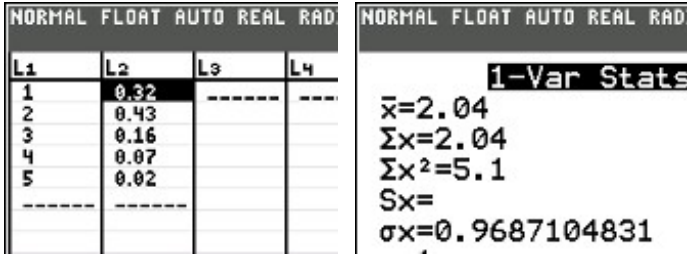
The policeman randomly stops 10 cars to inspect the passengers at a particular roadblock, and the total number of passengers inspected is the random variable Y .

- (ii)** Find $E(Y)$ and $\text{Var}(Y)$. [2]

The policeman also checks the car boot if the number of passengers in the car is not more than 2.

- (iii)** Find the probability that he checks more than 5 car boots at that roadblock. [2]

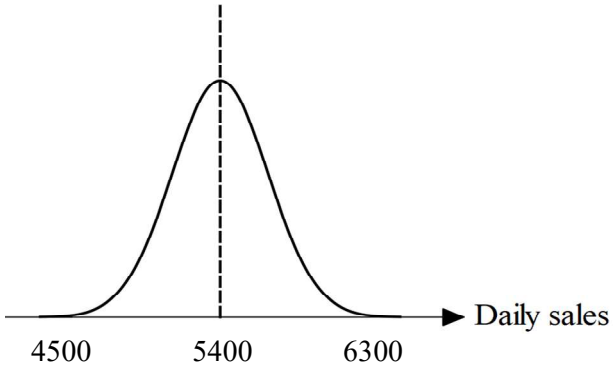
- (iv)** Find the probability that the last car is the 5th car boot that he checks at that roadblock. [2]

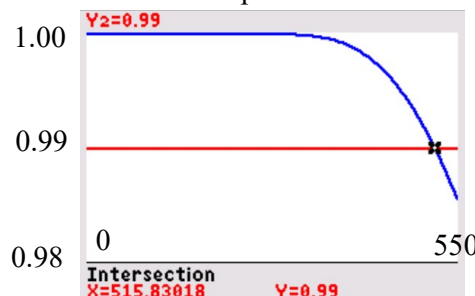
Solution		Comments
(i) [2]	$k = 1 - 0.32 - 0.43 - 0.16 - 0.07 = 0.02$ $E(X) = \sum x P(X = x) = 2.04$	
(ii) [2]	$Y = X_1 + X_2 + \dots + X_{10}$ $E(Y) = 10E(X) = 20.4$  $\text{Var}(X) = E(X^2) - (E(X))^2 = 0.96871^2 = 0.9384$ $\text{Var}(Y) = 10\text{Var}(X) = 9.384$	A few wrote • $\text{Var}(X) = E(X^2)$ • $\text{Var}(Y) = 10^2 \text{Var}(X)$ It is stated on the cover page that “Give non-exact numerical answers correct to 3 significant figures”. So, $\text{Var}(Y) = 9.384$ is an exact answer, there is no need to correct to 3 s.f.
(iii) [2]	Let C be the number of car boots checked by the police, out of 10 cars. Then, $C \sim B(10, 0.32 + 0.43 = 0.75)$ $P(C > 5) = 1 - P(C \leq 5) = 0.922$ (3.s.f.)	Common mistake $P(C > 5) = 1 - P(C \leq 4)$

(iv) [2]	Let D be the number of car boots checked by the police, out of 9 cars. Then, $D \sim B(9, 0.75)$ The required probability = $P(D = 4) \times 0.75$ $= 0.0389328 \times 0.75$ $= 0.0292$ (3 s.f.)	
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Source: 2022 Y6CT Q12

11	In this question you should state the parameters of any distributions that you use. A petrol station (Station A) finds that its daily sales, in litres, are normally distributed with mean 5400 and standard deviation 240.	
(i)	Sketch the distribution for the daily sales of petrol at Station A between 4500 litres and 6300 litres. [2]	
(ii)	Find the expected number of days of the year (365 days) the daily sales of petrol at Station A exceed 5950 litres. [2]	
(iii)	Find the probability that the total daily sales of petrol at Station A on 6 randomly chosen days is less than 32000 litres. [2]	
	The daily sales at another petrol station (Station B) are normally distributed with mean 6200 litres and standard deviation s litres.	
(iv)	The probability that daily sales of petrol at Station B on a randomly chosen day are more than 5000 litres is 0.99. Find the value of s . [3]	
(v)	Find the probability that, on a randomly chosen day, the daily sales of petrol on at Station B are more than half of that at Station A by at least 2000 litres. State an assumption needed for your calculation. [3]	

Solution		Comments
(i) [2]		Most are able to draw the bell-shaped curve symmetrically about 5400. The probability should be approximately zero for more than 3 standard deviations from the mean – in this case below 4680 and above 6120. Thus, at 4500 and 6300, the curve should be very very close to the horizontal axis with the

		curve roughly horizontal at these points. See Chap S3 notes page 4. Similar question has appeared in recent A-level examination.																																																
(ii) [2]	Let X be the daily sales of petrol at station A . Then $X \sim N(5400, 240^2)$ The expected number of days is $365 P(X > 5950) \approx 4.001 = 4.00$ (3 s.f.)																																																	
(iii) [2]	Let $W = X_1 + X_2 + X_3 + X_4 + X_5 + X_6$ $E(W) = 6E(X) = 6 \times 5400 = 32\,400$ $\text{Var}(W) = 6\text{Var}(X) = 6 \times 240^2 = 345\,600$ $W \sim N(32\,400, 345\,600)$ $P(W < 32000) = 0.248$ (3 s.f.)	Common mistake: some used 345600 as the standard deviation when keying into the GC, instead of $\sqrt{345600}$																																																
(iv) [3]	Method 1 Let Y be the daily sales of petrol at station B . Then $Y \sim N(6200, s^2)$ $P(Y > 5000) = 0.99$ $P\left(Z > \frac{5000 - 6200}{s}\right) = 0.99$ $P\left(Z > -\frac{1200}{s}\right) = 0.99$ From GC, $P(Z > -2.32635) = 0.99$ Thus, $-\frac{1200}{s} = -2.32635$ $s = 515.83$ (5 s.f.) $= 516$ (3 s.f.) Method 2 We can use GC to draw the graph of $y = \text{normalcdf}(500, E99, 6200, x)$ and $y = 0.99$, to see where they intersect. The x -coordinate of the point of intersection would represent the value of s . 	Some found the value of s by using a table of values. However, this method is not suitable for this question normalcdf lower: 5000 upper: E99 μ: 6200 σ: X Paste Plot1 Plot2 Plot3 $\text{Y1} = \text{normalcdf}(5000, E99,$<table><tr><th>X</th><th>Y1</th><th></th><th></th></tr><tr><td>507</td><td>0.991</td><td></td><td></td></tr><tr><td>508</td><td>0.9909</td><td></td><td></td></tr><tr><td>509</td><td>0.9908</td><td></td><td></td></tr><tr><td>510</td><td>0.9907</td><td></td><td></td></tr><tr><td>511</td><td>0.9906</td><td></td><td></td></tr><tr><td>512</td><td>0.9905</td><td></td><td></td></tr><tr><td>513</td><td>0.9903</td><td></td><td></td></tr><tr><td>514</td><td>0.9902</td><td></td><td></td></tr><tr><td>515</td><td>0.9901</td><td></td><td></td></tr><tr><td>516</td><td>0.99</td><td></td><td></td></tr><tr><td>517</td><td>0.9899</td><td></td><td></td></tr></table>Y1=0.98997957695137 Table of values method should be used when the answer is an integer. What if the answer is $s = 2.13$ (3 s.f.)? From the table above, it seems like $s = 516$ gives the probability to be exactly 0.99. However,	X	Y1			507	0.991			508	0.9909			509	0.9908			510	0.9907			511	0.9906			512	0.9905			513	0.9903			514	0.9902			515	0.9901			516	0.99			517	0.9899		
X	Y1																																																	
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	From GC, $s = 515.83$ (5 s.f.) $= 516$ (3 s.f.) Note: this method is not encouraged because there is not much working. If you do not get your answer correctly, you may not get any marks at all as no method is seen.	when you move the cursor to the value, it is actually 0.98997... Because the space in the table can only show up to 5 digits, when rounded off to 4 d.p., it shows 0.99.
(v) [3]	Let $T = Y - \frac{1}{2}X$ $E(T) = E(Y) - \frac{1}{2}E(X) = 6200 - \frac{1}{2} \times 5400 = 3500$ $\text{Var}(T) = \text{Var}(Y) + \left(\frac{1}{2}\right)^2 \text{Var}(X)$ $= 515.83^2 + \frac{1}{4} \times 240^2 = 280480.6$ $T \sim N(3500, 280480.6)$ $P(T \geq 2000) = 0.998 \text{ (3 s.f.)}$ Assumption: The daily sales of petrol at Station A is independent of the daily sales at Station B.	Some used the value of s to be 516, and this gives an over-estimated value for the variance of T. Some gave $\text{Var}(T) = 515.83^2 - \frac{1}{4} \times 240^2$ Quite a number of scripts missed the last part of the question, which is to state the assumption

Source : 2020 Y6 CT2 Q12

12	In this question you should state clearly all the distributions that you use, together with the values of the appropriate parameters. A local supermarket sells two types of potatoes, Russet potatoes and Holland potatoes. The masses, in grams, of the Russet potatoes have the distribution $N(200, 30^2)$ and the masses, in grams, of the Holland potatoes have the distribution $N(150, 24^2)$.
(i)	Find the probability that the total mass of 3 randomly chosen Holland potatoes is less than twice the mass of a randomly chosen Russet potato. [3]
(ii)	The supermarket decides to pack the potatoes into a variety pack. Each variety pack consists of 5 randomly chosen Russet and 4 randomly chosen Holland potatoes. The probability that a randomly chosen variety pack is within k grams of 1600 grams is found to be 0.775. Find k . [3]
(iii)	40 variety packs are randomly selected and are to be donated to needy families. Using the value of k found in (ii), find the probability that at most 5 variety packs are not within k grams of 1600 grams, [2]
(iv)	State an assumption needed for your calculations in parts (i), (ii) and (iii). [1]

	The supermarket recently introduced a new type of potatoes, called New potatoes. The mean mass of New potatoes is 55 grams and standard deviation is 13 grams.	
	(v)	Find an approximate value for the probability that the average mass from a random sample of 100 New potatoes is not more than 58 grams. [3]
Solution		
(i)	Let R and H be the mass, in grams, of a Russet and Holland potato respectively. Then $R \sim N(200, 30^2)$ and $H \sim N(150, 24^2)$ Let $T = H_1 + H_2 + H_3 - 2R$ $E(T) = 3E(H) - 2E(R) = 3 \times 150 - 2 \times 200 = 50$ $\text{Var}(T) = 3\text{Var}(H) + 2^2 \text{Var}(R) = 3 \times 24^2 + 2^2 \times 30^2 = 5328$ $T \sim N(50, 5328)$ $P(H_1 + H_2 + H_3 < 2R)$ $= P(H_1 + H_2 + H_3 - 2R < 0)$ $= 0.247$ (3s.f.)	Define random variable clearly. Do NOT use “Z” which is for $N(0,1)$ Indicate the distribution clearly in ALL parts.
(ii)	Let $V = R_1 + R_2 + \dots + R_5 + H_1 + H_2 + \dots + H_4$ $E(V) = 5E(R) + 4E(H) = 5 \times 200 + 4 \times 150 = 1600$ $\text{Var}(V) = 5\text{Var}(R) + 4\text{Var}(H) = 5 \times 30^2 + 4 \times 24^2 = 6804$ $V \sim N(1600, 6804)$ $P(1600 - k < V < 1600 + k) = 0.775$ $k = 100$ (3s.f.)	
(iii)	Let W be the number of variety packs not within “ k ” grams of the mean out of 40 packs. $W \sim B(40, 1 - 0.775 = 0.225)$ $P(W \leq 5) = 0.0869$ (3sf)	Define the binomial random variable clearly
(iv)	The mass of a potato is independent of the mass of another potato. Please note: “Distributions” or “Probabilities” CANNOT be described as being independent of each other. “Events” or “Random Variables” can be described as independent.	
(v)	Let C be the mass of a New potato. $E(C) = 55$ and $\text{Var}(C) = 13^2$ Since $n = 100$ is large, by Central Limit Theorem, $\bar{C} \sim N\left(55, \frac{13^2}{100}\right)$ approximately $P(\bar{C} \leq 58) = 0.989$ (to 3s.f.)	You cannot claim that C is normally distributed as that is not specified. Notice also that the question asks to “approximate” the probability.